Algebra 2 Name: KEY

Unit 6B Practice Test Period: _____

Show all work for full credit ☺

Solve for x. Box Answers

1) $\sqrt{4x-5} + 10 = 17$ Isolate then Square both sides

$(\sqrt{4x-5})^2 = (7)^2$

$4x-5 = 49$

$4x = 54$

$x = \frac{54}{4} = \frac{\cancel{2}(27)}{\cancel{2}(2)}$

$x = \frac{27}{2}$

2) $3\sqrt{x+7} - 8 = 2$

$3\sqrt{x+7} = 10$

$\sqrt{x+7} = \frac{10}{3}$

$x+7 = \frac{100}{9}$

$x = \frac{100}{9} - 7$

$x = \frac{100}{9} - \frac{7}{1}\left(\frac{9}{9}\right)$

$= \frac{100}{9} - \frac{63}{9}$

$= \frac{37}{9} \approx 4.1$

EXACT APPROXIMATE

make it easier

$3\sqrt{x+7} - 10 = 2$

$3\sqrt{x+7} = 12$

$\sqrt{x+7} = 4$

$x+7 = 16$

$x = 9$

3) $(\sqrt{2x+3})^2 = (x)^2$

$2x+3 = x^2$

$0 = x^2 - 2x - 3$

$= (x-3)(x+1)$

$\boxed{x = 3}$ $\cancel{x = -1}$

Check

$\sqrt{2(3)+3} = 3$ $\sqrt{2(-1)+3} \stackrel{?}{=} -1$

$\sqrt{9} = 3$

✓

4) $(\sqrt{24-5x})^2 = (x-6)^2$

$24 - 5x = x^2 - 12x + 36$

$+5x \quad +5x$

$0 = x^2 - 7x + 12$

$0 = (x-3)(x-4)$

$\cancel{x = 3}$ $\cancel{x = 4}$

Check

$\sqrt{24-5(3)} = 3-6$

$\sqrt{9} = -3$

$\sqrt{24-5(4)} = 4-6$

$\sqrt{24-20} = -2$

$\sqrt{4} \neq -2$

$2 \neq -2$

Find the domain of each function. Answer in **Interval Notation**

5) $f(x) = \sqrt{3x + 42}$ $3x+42 \geq 0$ $3x \geq -42$ $x \geq 14$ $[14, \infty)$ $3\overline{)42}$ 14, 3, 12	6) $v(x) = \frac{5}{7}x^2 - 8$ $(-\infty, \infty)$
7) $g(x) = \frac{x-9}{2x+1}$ $2x+1 \neq 0$ $2x \neq -1$ $x \neq -\frac{1}{2}$ $(-\infty, -\frac{1}{2})(-\frac{1}{2}, \infty)$	8) $h(x) = \frac{x+11}{x^2+7x+12}$ $(x+3)(x+4) \neq 0$ $x+3 \neq 0$ $x+4 \neq 0$ $x \neq -3$ $x \neq -4$ $(-\infty, -4)(-4, -3)(-3, \infty)$

Perform the indicated function operation.

9) If $f(x) = x^2 + 3$ and $g(x) = 2x - 5$, find $f(g(-2))$. $g(-2) = 2(-2)-5 = -4-5 = -9$ $f(-9) = (-9)^2 + 3$ $= 81 + 3$ $= 84$	10) If $f(x) = 5x + 1$ and $g(x) = -3x^2 + 2$ find $(g \circ f)(-5)$. $g(f(-5))$ $f(-5) = 5(-5)+1 = -25+1 = -24$ $g(-24) = -3(-24)^2 + 2$ $= -1728 + 2$ $= -1726$
11) If $f(x) = x^2 + 3$ and $g(x) = 2x - 5$, find: $(f - g)(x)$. $x^2 + 3 - (2x - 5)$ $x^2 + 3 - 2x + 5$ $x^2 - 2x + 8$	12) If $f(x) = 5x + 1$ and $g(x) = -3x^2 + 2$, find: $(f + g)(x)$. $= 5x+1 + -3x^2 + 2$ $-3x^2 + 5x + 3$

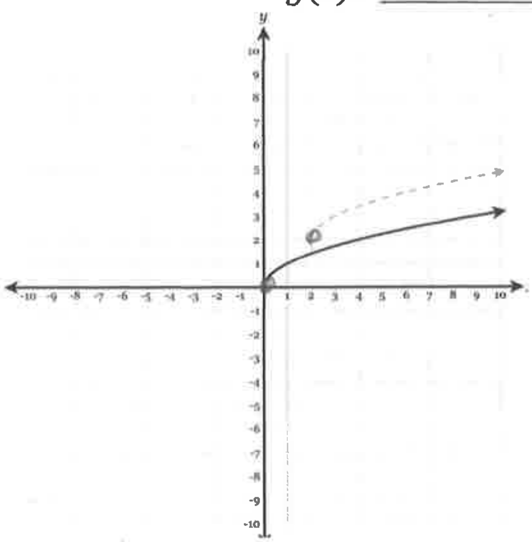
Problems involving inverse functions. Use proper notation

13) Find the inverse of $\cancel{f(x)} = x^{\frac{4}{5}}$ y $x = y^{\frac{4}{5}}$ $(x)^{\frac{5}{4}} = \left(y^{\frac{4}{5}}\right)^{\frac{5}{4}}$ $x^{\frac{5}{4}} = y$ $f^{-1}(x) = x^{\frac{5}{4}}$	14) Find the inverse of $\cancel{f(x)} = \sqrt[3]{x} + 2$ y $x = \sqrt[3]{y} + 2$ $x - 2 = \sqrt[3]{y}$ $(x-2)^3 = \left(\sqrt[3]{y}\right)^3$ $y = (x-2)^3$ $f^{-1}(x) = (x-2)^3$
15) Find the inverse of $\cancel{f(x)} = \frac{x}{x+3}$ y $x = \frac{y}{y+3}$ mult by $(y+3)$ $(y+3)(x) = y$ distribute $xy + 3x = y$ Gather y terms on one side $xy - y = -3x$ $y(x-1) = -3x$ factor out y $y = \frac{-3x}{x-1}$ divide by $(x-1)$ $f^{-1}(x) = \frac{-3x}{x-1}$ inverse notation OR $\frac{3x}{1-x}$	16) Evaluate: (A) $h^{-1}(h(15))$ 15 (B) $k(k^{-1}(4x))$ $4x$

Graphing radical equations.

17) The graph of $y = f(x)$ is the solid graph below. What is the function, $g(x)$, of the dotted graph?

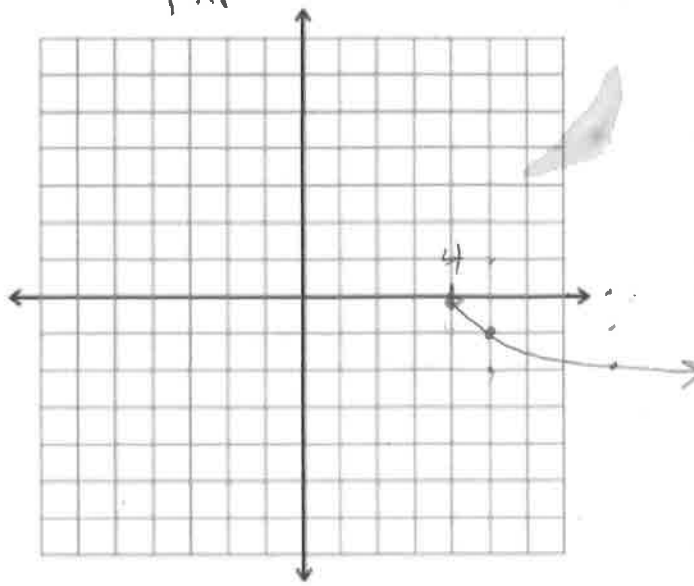
$g(x) =$ $\sqrt{x-2} + 2$



right 2
up 2

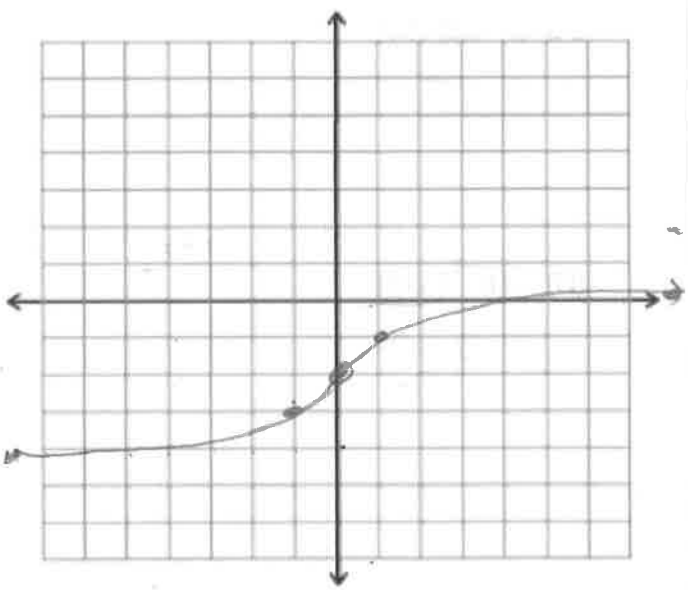
18) Graph $f(x) = -\sqrt{x-4}$

flip
right 4



19) Graph $f(x) = \sqrt[3]{x} - 2$

Down 2



20) Graph $f(x) = \sqrt[3]{x+1} + 4$

left 1 up 4